

During infection with a microbial pathogen, which of the following would most likely be responsible for phagocytosis of organisms and the release of histamine, respectively?

- ☐ A. Basophils, macrophages [3%]
- ☐ B. Natural killer cells, T cells [6%]
- ✓ ☒ C. Macrophages, basophils [87%]
- ☐ D. T cells, natural killer cells [2%]

Correct

87% answered correctly

Explanation:

Cells of the immune system	
Cell type	Primary function
B cells	Identify foreign pathogens, present antigen & produce antibodies
T cells	Mediate a number of immune responses, including those against viruses, bacteria & parasites
Dendritic cells	Identify foreign pathogens, present antigen & activate other immune cells
Macrophages	Degrade pathogens & dead body cells via phagocytosis
Basophils	Release chemical mediators such as histamine that enhance an immune response
Mast cells	Release chemical mediators such as histamine that modulate allergic reactions
Neutrophils	Kill & phagocytize bacterial cells
Eosinophils	Defend against parasitic infections & modulate immune responses during allergic reactions
Natural killer cells	Release toxins to destroy virus-infected body cells

A number of cells and proteins are directly involved in protecting the body against **foreign pathogens** (immunity). Some of these cells and proteins, together with the skin and mucous membranes, constitute the **innate immune system**, a network of **nonspecific defenses** acting to prevent or respond rapidly (minutes to hours) to infection by invading pathogens. The cells that make up the innate immune system include **macrophages**, dendritic cells, natural killer cells, and basophils.

The primary function of macrophages, which can either travel the body or remain fixed in a specific tissue, is to degrade pathogens and dead body cells. The degradative function of macrophages is performed via **phagocytosis**, the process of **engulfing solid particles** (eg, microbial pathogens) designated for destruction into phagocytic vesicles called phagosomes. Within the cell, phagosomes typically fuse with lysosomes to form **phagolysosomes**, which enable the degradation of the engulfed particles using specialized enzymes that work at **low pH**.

In contrast, the primary function of **basophils**, a type of granulocyte, is to release chemical mediators such as **histamine** that enhance the immune response. Released histamine causes **vasodilation** and **permeability** of vessels, which stimulates an inflammatory response by

increasing blood flow and the delivery of immune cells (eg, neutrophils and macrophages) at the site of infection. Therefore, during an infection, macrophages function to phagocytose organisms and basophils release histamine.

(Choice A) The question asks for the immune cells most likely responsible for phagocytosis and histamine release, respectively (ie, in that order), but the answer choices here are reversed. Although it has been noted that basophils can perform phagocytosis and macrophages can release histamine, these processes are uncommon. As a result, the reverse order (macrophages, basophils) is the *best* answer choice.

(Choices B and D) **Natural killer cells** are cellular components of the innate immune system. Natural killer cells release toxins to destroy body cells, such as neoplastic and virus-infected cells, that have reduced expression of major histocompatibility complex molecules. T cells are cellular components of the **adaptive immune system**. They are important in mediating a number of immune responses, including those against viruses. These types of cells are not generally known to directly phagocytose bacteria during an immune response.

Educational objective:

Macrophages are one of the cells responsible for phagocytosis, the process of engulfing solid particles (eg, bacteria) designated for destruction into phagocytic vesicles called phagosomes. In contrast, basophils function to release chemical mediators such as histamine that stimulate an inflammatory response.

Time spent:0 sec

Subject:
Biology

QID:401296

System:
Skin and Immune Systems

Which of the following cell-cell junctions would help skin cells withstand mechanical stress by directly anchoring the intermediate filaments of adjacent cells?

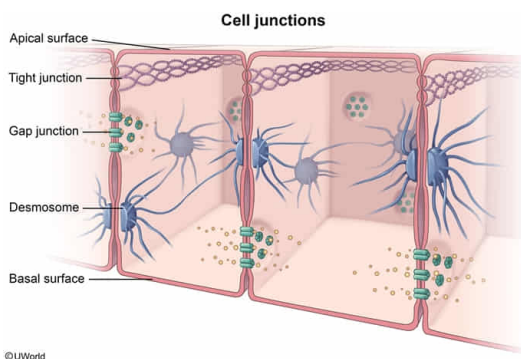
- I. Desmosomes
- II. Gap junctions
- III. Tight junctions

- ✓ ☒ A. I only [36%]
- ☐ B. II only [4%]
- ☐ C. I and III only [46%]
- ☐ D. II and III only [11%]

Correct

37% answered correctly

Explanation:



Cell-cell junctions, such as desmosomes, gap junctions, and tight junctions, are structures that attach adjacent cells to one another. Cell junctions are prominent in epithelial tissue, which is composed of densely packed cells and very little extracellular matrix.

Desmosomes provide **tensile strength** to epithelial cell sheets by **anchoring the cytoskeletons**, specifically the intermediate filaments, of two cells together. This creates a continuous cytoskeletal network that spans the entire epithelial sheet, one through which mechanical stress (eg, pulling, stretching, tension) can be distributed. Desmosomes are found predominantly in tissues that are subject to high levels of mechanical stress (eg, muscle tissue, epithelial layers of **the skin**) and prevent tissue tearing (**Number I**).

(Number II) Gap junctions are cell-cell junctions that *mediate communication* between cells. Protein channels (**connexons**) in one cell align with complementary channels in another cell to form pores that facilitate the passive and bidirectional exchange of ions and small solutes. Gap junctions are found in cell populations that depend on coordinated activity, such as smooth muscle, cardiac muscle, or neural tissue.

(Number III) Tight junctions are cell-cell junctions that *prevent water and solutes from diffusing* between cells and across the epithelial cell layer. These junctions form a watertight

seal that fully encircles the apical end of every cell in the basal epithelial sheet. Tight junctions serve as a barrier and separate tissue space; they are found in a number of tissues, including skin, gastrointestinal tract, and testis.

Educational objective:

Desmosomes, gap junctions, and tight junctions are cell-cell junctions that provide tissue with mechanical strength, cytoplasmic continuity, and watertight seals, respectively.

Time spent:0 sec

Subject:
Biology

QID:401297

System:
Skin and Immune Systems